

判斷鏡像串

鏡像串就是左右鏡像之後和原資料相同，如 2S、3AIAE、bmpqmd。並不是每個字元在鏡像之後都能得到一個合法字元。在本題中，每個字元的鏡像如下表所示(斜線表示該字元鏡像後不能得到一個合法字元)。

Character	Reverse	Character	Reverse	Character	Reverse	Character	Reverse	Character	Reverse
A	A	a	/	N	/	n	n	0	0
B	/	b	d	O	O	o	o	1	1
C	/	c	/	P	/	p	q	2	S
D	/	d	b	Q	/	q	p	3	E
E	3	e	/	R	/	r	/	4	/
F	/	f	/	S	2	s	/	5	Z
G	/	g	/	T	T	t	/	6	/
H	H	h	/	U	U	u	/	7	/
I	I	i	i	V	V	v	v	8	8
J	L	j	/	W	W	w	w	9	/
K	/	k	/	X	X	x	x		
L	J	l	l	Y	Y	y	/		
M	M	m	m	Z	5	z	/		

試撰寫程式去判斷輸入字串是否為鏡像串，按照下列規則輸出對應文字：

- 1) 若是鏡像串，則輸出該字串後加上"-- is a mirrored string."
- 2) 若不是鏡像串，則輸出該字串後加上"-- is only a string."

輸入說明：

第 1 行為 L 筆欲判斷字串數，其範圍為 $1 \leq L \leq 100$ 。

第 2 行至第 1+L 行輸入字串 P，其中 $P \in \{A-Z, a-z, 0-9\}$ ，字串長度 n 的範圍為 $2 \leq n \leq 50$ 。

輸出說明：

依據每行輸入字串內容，判斷該字串是否為鏡像串，並按照上述規則輸出對應文字

範例說明：

範例 1 輸入	範例 1 輸出
2 2A3MEAS NOTAPALINDROME	2A3MEAS -- is a mirrored string. NOTAPALINDROME -- is only a string.
範例 2 輸入	範例 2 輸出
2 aCD7DCa AmTOvYvOTmA	aCD7DCa -- is only a string. AmTOvYvOTmA -- is a mirrored string.

Mirrored Strings

A mirrored string is a string that remains identical to the original data after being mirrored horizontally, such as "2S", "3AIAE", and "bmpqmd". Not every character yields a valid character after being mirrored. In this problem, the mirrored character for each character is shown in the table below (a slash indicates that the character does not yield a valid character after being mirrored).

Character	Reverse	Character	Reverse	Character	Reverse	Character	Reverse	Character	Reverse
A	A	a	/	N	/	n	n	0	0
B	/	b	d	O	O	o	o	1	1
C	/	c	/	P	/	p	q	2	S
D	/	d	b	Q	/	q	p	3	E
E	3	e	/	R	/	r	/	4	/
F	/	f	/	S	2	s	/	5	Z
G	/	g	/	T	T	t	/	6	/
H	H	h	/	U	U	u	/	7	/
I	I	i	i	V	V	v	v	8	8
J	L	j	/	W	W	w	w	9	/
K	/	k	/	X	X	x	x		
L	J	l	l	Y	Y	y	/		
M	M	m	m	Z	5	z	/		

Write a program to determine whether the input string is a mirrored string, and output the corresponding text according to the following rules:

1. If it is a mirrored string, output the string followed by "-- is a mirrored string."
2. If it is not a mirrored string, output the string followed by "-- is only a string."

Input Description:

The 1st line inputs L, the number of strings to be determined, where the range is $1 \leq L \leq 100$. The 2nd line to the (1+L)-th line input the string P, where $P \in \{A-Z, a-z, 0-9\}$, and the string length n ranges from $2 \leq n \leq 50$.

Output Description:

Based on the content of each input string, determine whether the string is a mirrored string, and output the corresponding text according to the rules mentioned above.

Example Description:

Example 1

Input

2

2A3MEAS

NOTAPALINDROME

Output

2A3MEAS -- is a mirrored string.

NOTAPALINDROME -- is only a string.

Example 2

Input

2

aCD7DCa

AmTOvYvOTmA

Output

aCD7DCa -- is only a string.

AmTOvYvOTmA -- is a mirrored string.